

Lactate Dehydrogenase (LDH) Inhibition Assay

Laboratory Report — Enzyme Kinetics and Dose-Response Analysis

Analyst: Gontse Bridgett Baloyi

Technique category: Enzyme Kinetics / Biochemistry

1. Objective

To measure the inhibitory effect of a candidate compound on Lactate Dehydrogenase (LDH) activity across a range of concentrations, and to estimate the IC₅₀ (the concentration producing 50% inhibition) by fitting a dose-response curve.

2. Introduction

LDH catalyses the reversible conversion of pyruvate to lactate, coupled with oxidation of NADH to NAD⁺. Because NADH absorbs strongly at 340 nm while NAD⁺ does not, the rate of decrease in absorbance at 340 nm is directly proportional to enzyme activity. By pre-incubating LDH with increasing concentrations of a candidate inhibitor and comparing the resulting reaction rate to an uninhibited control, percent inhibition can be calculated at each dose, and the overall potency of the inhibitor summarised as an IC₅₀ value.

3. Materials

- Purified LDH enzyme (or lysate with LDH activity)
- Substrate solution: sodium pyruvate + NADH in phosphate buffer (pH 7.4)
- Candidate inhibitor, serially diluted across a concentration range
- UV-transparent microplate or cuvettes
- Spectrophotometer / microplate reader (340 nm)

4. Method

- Prepared an 8-point, 2-fold serial dilution series of the candidate inhibitor.
- Pre-incubated LDH with each inhibitor concentration for 5–10 minutes.
- Initiated the reaction by adding substrate (pyruvate + NADH).
- Recorded absorbance at 340 nm at regular intervals over 2–5 minutes.
- Calculated the initial reaction rate (slope of A₃₄₀ vs. time) for each concentration.
- Calculated percent inhibition relative to the uninhibited control.
- Fitted a sigmoidal dose-response curve to estimate IC₅₀.

5. Results

Reaction rate decreased progressively with increasing inhibitor concentration, from 0.182 A₃₄₀/min in the uninhibited control to 0.012 A₃₄₀/min at the highest concentration tested (1000 μM), corresponding to 93.4% inhibition.

Inhibitor (uM)	Rate (A340/min)	% Inhibition
0.0	0.182	0.0
0.1	0.178	2.2
1.0	0.151	17.0
10.0	0.098	46.2
100.0	0.041	77.5
1000.0	0.012	93.4

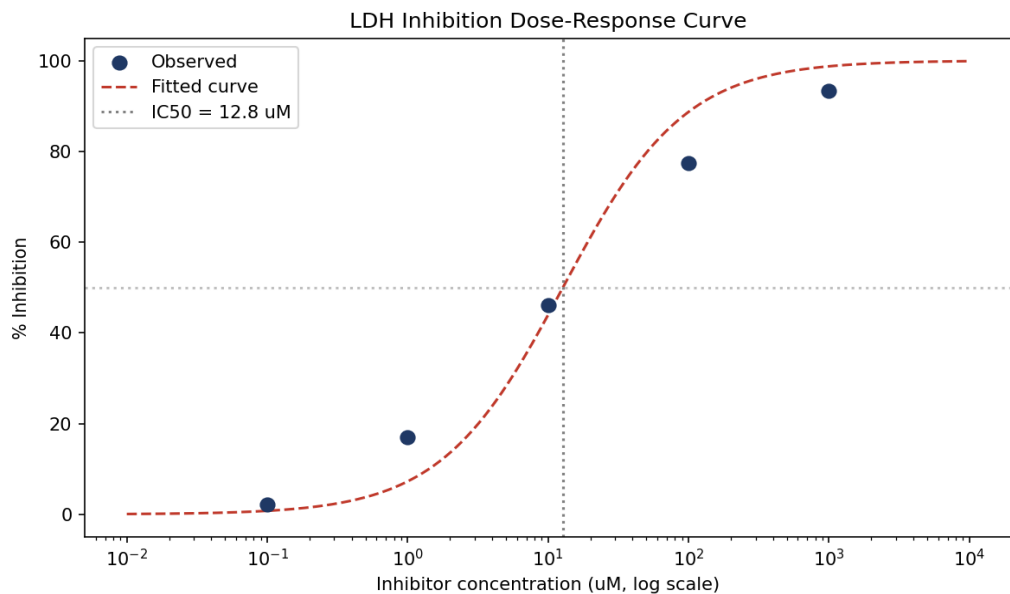


Figure 1. LDH inhibition dose-response curve with fitted sigmoidal model. $IC_{50} = 12.8 \text{ uM}$.

The fitted dose-response curve gave an estimated IC_{50} of approximately **12.8 uM**, the concentration at which the inhibitor reduced LDH activity by half.

6. Discussion

The dose-response relationship followed the expected sigmoidal shape, with inhibition increasing steeply across roughly two orders of magnitude of inhibitor concentration (1–100 uM) before plateauing near-complete inhibition. This pattern is consistent with a single-site, reversible inhibition mechanism. The calculated IC_{50} provides a standard potency metric that could be used to rank this inhibitor against other candidate compounds tested under the same assay conditions. Because IC_{50} is assay-dependent (affected by substrate concentration and enzyme concentration, per the Cheng-Prusoff relationship), comparisons across different assay conditions would require normalisation.

7. Conclusion

This exercise demonstrated a complete enzyme inhibition workflow, from raw kinetic absorbance data through percent-inhibition calculation to dose-response curve fitting and IC_{50} estimation — a standard approach used in enzyme inhibitor screening and characterisation.